

A Theoretical Framework

In this section, we introduce a theoretical framework to guide our empirical analysis.

In Section A.1, we present a simple partial-equilibrium model of chatbot adoption to interpret the descriptive patterns in Section 2.2. We posit a Roy (1951)-style selection model in which employer initiatives shift both the costs and benefits of adoption, and analyze how these shifts affect workers' adoption decisions and the work-related benefits they report.

In Section A.2, we embed this adoption model within a labor market equilibrium framework in which differentiated firms adopt chatbot initiatives and hire workers to maximize profits. We use this model to guide and interpret the empirical analysis of labor market outcomes in Section 3.

A.1 Adoption and Work

A.1.1 Setup

Worker i derives benefits B_i from using AI chatbots:

$$B_i = \alpha_b + \beta_b \times i, \quad (4)$$

where individuals $i \sim \mathcal{U}([0, 1])$ are ordered by their benefits, such that $\beta_b \leq 0$. While we assume linear functional forms for analytical simplicity, our predictions hold under more general monotonicity assumptions.

The worker also incurs a cost C_i of adopting chatbots:

$$C_i = \alpha_c + \beta_c \times i, \quad (5)$$

where $\beta_c \geq 0$ if workers who benefit more from AI also face lower adoption costs.²⁴

We conceptualize employer initiatives $E \in \{0, 1\}$ as shifting workers' costs and benefits

²⁴A common finding from RCTs is that AI chatbots yield greater benefits for less experienced (and thus typically younger) workers (Brynjolfsson, Li and Raymond, 2025; Dell'Acqua et al., 2024; Noy and Zhang, 2023). If these workers also find it easier to adopt new tools, then $\beta_c \geq 0$.

through the parameters α_c and α_b :

$$\alpha = \alpha_0 + \alpha_1 E. \quad (6)$$

While we focus on a binary initiative for analytical simplicity, our propositions extend naturally to combinations of initiatives—such as use policies, enterprise chatbots, and training.

Workers adopt AI chatbots if their perceived benefits exceed the associated costs ($B_i \geq C_i$):

$$A_{ij} = \mathbf{1}[B_{ij} \geq C_{ij}] = \mathbf{1}[\hat{\alpha}_0 + \hat{\alpha}_1 E_j + \hat{\beta} \times i \geq 0], \quad (7)$$

where $\hat{x} = x_b - x_c$ denotes net benefits.

A.1.2 Model Predictions

Figure A.1 illustrates how employer initiatives shift adoption rates and average reported benefits; formal derivations follow below.

Adoption rates. Table 1, Columns 1–2 show how chatbot adoption rates vary with employer initiatives. The optimal adoption rate is:

$$i^*(E) = \mathbf{P}[B_{ij} \geq C_{ij}] = -\frac{\hat{\alpha}}{\hat{\beta}}, \quad (8)$$

where $0 < \hat{\alpha} < -\hat{\beta}$, corresponding to the empirically relevant case in which adoption rates lie between 0 and 1.

Equation (8) implies that initiatives raising net benefits ($\hat{\alpha}$) increase adoption, with stronger effects when $\hat{\beta}$ is numerically small—i.e., when workers are relatively homogeneous in their net benefits from AI chatbots.

In this light, the fact that employer-encouraged use in Table 1 is associated with a *doubling* of adoption rates suggests that encouragement is powerful relative to the degree of individual heterogeneity in net benefits.

Benefits for users. Table 1, Columns 3–6 show how reported benefits among chatbot users vary with employer initiatives. The average benefit reported by users is:

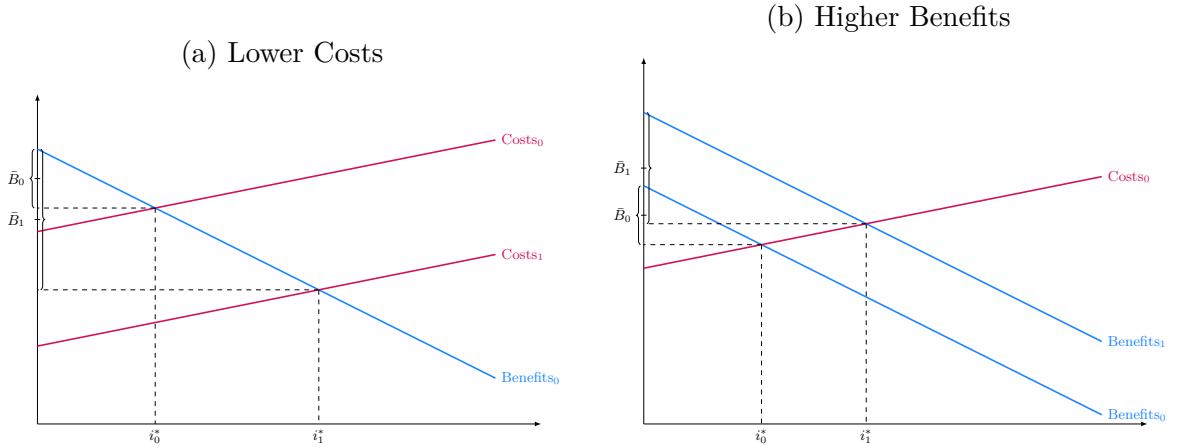
$$\bar{B}(E) = \mathbb{E}[B_i(\alpha) \mid i \leq i^*(\alpha)] = \gamma\alpha_c + (1 - \gamma)\alpha_b, \quad (9)$$

where $\gamma = \frac{\beta_b}{2\beta} \geq 0$. Equation (9) yields two key insights.

First, initiatives that reduce adoption costs (α_c) *lower* average reported benefits through a *selection effect*: lower costs induce adoption by workers with lower idiosyncratic benefits (higher i). This negative selection effect is stronger when workers are more homogeneous in net benefits ($\hat{\beta}$ numerically small, so initiatives generate larger adoption increases; cf. Equation (8)) but more heterogeneous in gross benefits (β_b large in absolute value, so marginal adopters have substantially lower benefits).

Second, initiatives that increase benefits (α_b) have two offsetting effects. A higher α_b directly raises each user's benefit B_i , but also expands adoption to workers with lower idiosyncratic benefits, triggering the same negative selection effect. The net effect on $\bar{B}$ is positive if and only if $\gamma < 1$ —a condition more likely when workers are similar in gross benefits (β_b small) but differ in adoption costs (β_c large).

Figure A.1: The Impact of Employer Initiatives on Adoption Rates and Reported Benefits



Notes: This figure illustrates how employer initiatives affect adoption rates (i^*) and average reported benefits among users ($\bar{B}$). Panel (a) shows initiatives that reduce adoption costs (α_c); Panel (b) depicts initiatives that enhance benefits (α_b). Lower costs unambiguously increase adoption and reduce average reported benefits via negative selection. Initiatives that raise benefits also increase adoption, but their impact on average reported benefits is theoretically ambiguous and, at most, equal to their individual-level effect.

In summary, average reported benefits among users provide a lower bound on the causal effect of employer initiatives on *individual* benefits, due to negative *selection*.

The fact that Table 1 shows higher reported benefits in encouraged settings is therefore striking. It implies two things: (i) employer encouragement primarily raises perceived benefits (α_b), since reductions in adoption costs alone would generate only the negative selection effect; and (ii) these benefit increases outweigh the adverse selection—implying $\gamma < 1$, so that encouragement is powerful relative to individual heterogeneity.

By contrast, enterprise chatbots and training—when implemented in isolation—are associated with lower reported benefits, potentially reflecting either that these initiatives reduce individual benefits ($\alpha_{b1} < 0$) or that negative selection effects dominate any positive gains.

A.1.3 Discussion

The preceding analysis examines the implications of workers self-selecting into chatbot adoption based on individual benefits. Beyond this margin, at least two additional forms of selection may be at play.

First, firms that encourage chatbots may be those with the greatest underlying benefits from the tools—even absent any initiatives—introducing potential reverse causality. In our framework, this corresponds to E being positively related to α_{b0} . Second, employers may target encouragement toward employees expected to benefit most individually, resembling a rotation of the cost and benefit schedules in Figure A.1 rather than simple vertical shifts.

Section F addresses both margins empirically. First, Section F.5 examines whether employers select into encouragement based on underlying benefits from chatbot use. We show that the effects of encouragement remain robust when controlling for firm- and worker-level characteristics, including firm age, size, and productivity, as well as workers’ detailed task mixes. Second, Section F.6 implements a “coworker encouragement” design, using the average encouragement reported by a worker’s colleagues to capture *workplace-*